

Agentic AI & Autonomous Multi-Agent Systems

Duration: 3 Months (12 Weeks) | Program Fee: ₹2,999 INR | Format: 100% Practical Labs & Capstone Defense

Program Overview: Design and deploy autonomous AI agents with reasoning loops, dynamic tool calling, stateful graphs in LangGraph, multi-agent swarms with CrewAI, and Model Context Protocol (MCP).

3-MONTH STRUCTURED ROADMAP & MODULES

Module 01: Agentic Architectures & Reasoning Loops

- ReAct (Reason + Act) loop architectures & Plan-and-Solve patterns
- Native function/tool calling with OpenAI, Claude & Ollama models
- Type-safe tool definitions, schema validation & error handling in Pydantic

Lab Practical: *Build an autonomous terminal agent capable of inspecting files, running commands and self-correcting errors.*

Module 02: Environment Execution & Secure Sandboxing

- REST API orchestration tools & dynamic OpenAPI spec calling
- SQL database querying agents with strict read-only security guards
- Browser automation with Playwright & containerized Docker sandbox execution

Architecture Pipeline: User Goal → Planner Agent → Tool Dispatcher → Docker Sandbox Runtime → Synthesizer Agent

Module 03: Stateful Graph Workflows with LangGraph

- Graph state machines (nodes, edges, conditional transitions & cycles)
- Checkpointers, persistent threads & time-travel debugging
- Human-in-the-loop approval breakpoints & state modification

Lab Practical: *Construct a stateful code-refactoring agent pipeline with human gate approval.*

Module 04: Multi-Agent Collaboration with CrewAI & AutoGen

- Hierarchical vs collaborative agent swarms & inter-agent delegation
- Role-playing specialized agents (Researcher, Architect, Coder, QA Tester)
- Shared memory contexts, conflict resolution & task dependency trees

Module 05: Model Context Protocol (MCP) & Long-Term Memory

- Anthropic Model Context Protocol (MCP) architecture & standard servers
- Building custom MCP servers to expose internal business systems to LLMs
- Episodic vs semantic long-term memory using Mem0 & vector graph storage

Lab Practical: *Develop a custom MCP server connecting external SQLite/Postgres databases to Claude Desktop & LangGraph.*

Module 06: Agent Observability, Reliability & Production Deployment

- Observability & tracing with Arize Phoenix, LangSmith & OpenTelemetry
- Infinite loop mitigation, timeout fallbacks & rate limit budgeting
- Production deployment with Docker, background task queues & streaming event websockets

CAPSTONE PROJECT DEFENSE & DELIVERABLES

Title: Autonomous Software Engineering & Research Swarm

Execution Pipeline: Feature Request → Project Manager Agent → Research Agent → Coding Agent in Sandbox → QA & Test Agent → Git Commit

Deliverables: Complete multi-agent codebase with LangGraph architecture, Live interactive dashboard with human-in-the-loop review, End-to-end trace telemetry logs demonstrating autonomous resolution

Official curriculum issued by Unisole Skill AI Labs. Direct employer verification available via LinkedIn and student dashboard.